Report I — SF2930 Regression Analysis

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Introduction

The World Health Organization (WHO) has reported that obesity is a major risk factor for a number of chronic diseases, including diabetes, cardiovascular diseases, and cancer. Obesity is defined as "the disease in which excess of body fat has accumulated to such extents that health may be adversely affected." Once considered a problem only for high-income countries, obesity is now rising in low- and middle-income countries. An important issue for medical purposes thus is to reliably identify people with excess fat.

The well-known body mass index (BMI), while widely used in practice and simple to calculate, is only an indirect measure of fatness and is empirically shown to be a poor predictor of actual fatness. Instead, a person's body fat mass (BFM) is considered. Highly accurate methods for measuring the BFM, such as X-ray densitometry (DXA) or hydrodensitometry, while being precise, have little practical applicability because of the high costs and methodological efforts. Thus, cheaper and more portable methods such as regression models attract a lot of interest in body composition research.

A number of anthropologic measurements such as waist circumference and waist-to-hip ratio, combined with skin-fold thickness are known to be related to the BFM. These variables can be used as predictor variables in multiple linear regression models, which can then be used for predicting BFM instead of measuring it exactly. For deeper understanding of the topic, see Garcia et al. (2005).

Dataset Description

This dataset contains measurements of the body density (density) of 252 men assessed by hydrodensitometry (underwater weighing) along with their age and a number of anthropometric variables. This is a modified version of the original dataset available in the R package *mfp*, with the columns *case* *brozek* and *siri* removed. Note that certain data points may be erroneous; see the data description at https://cran.r-project.org/web/packages/mfp/mfp.pdf. For a more detailed presentation of the dataset, see http://lib.stat.cmu.edu/datasets/bodyfat and Izenman (2009, Example 5.5.2).

Project goals

In this project, we will establish assessments regarding body fat in men. While scientific and precise methods exist to accurately determine a person's body fat, these methods are often expensive and hard to deploy. Therefore, by measuring certain body parameters and attempting to establish a relationship between these parameters and body density, we will try to identify a pattern to predict as accurately as possible the BFM. The dataset is composed of 252 measurements and 14 parameters such as: density, age, weight, height, neck, chest, abdomen, hip, thigh, knee, ankle, biceps, forearm and wrist. Consequently, the inequality $p < n$ is indeed verified. We can then try to make a multiple linear regression of the model, with density to be the predictive variable in the model.

Analysesand model development

In every analysis of data, we must first understand the nature of the parameters. Consequently, we have to see if they are well dimensioned, and if the dataset doesn't contain any NaaN value that might disturb the model.

0.1 Dealing with multicollinearity

We will then analyze the possible relations between the input parameters in order to detect any pattern between the variables. Indeed, a clear relation between one or more predictor in the matrix X might lead to the phenomenon of multicolinearity. This phenomenon creates numerical instability in the estimation of the regression coefficients and can lead to unreliable statistical inferences.

Mathematically, multicollinearity implies that the columns of $\mathbf{X}$ are linearly dependent or nearly linearly dependent, which means that there exists a set of nonzero constants $c_1, c_2, \ldots, c_p$ such that:

$$c_1\mathbf{x}_1 + c_2\mathbf{x}_2 + \cdots + c_p\mathbf{x}_p \approx 0.$$

where $\mathbf{x}_j$ represents the j-th column of $\mathbf{X}$.

To detect this, we first compute the correlation matrix. This matrix represents the values of the centered matrix $\mathbf{X}^T\mathbf{X}$. If the matrix has values off the diagonal close to 1 or -1, it means that the predictors associated might be correlated. However, the correlation matrix only gives possible relations between two predictors.

A more systematic way to detect multicollinearity is by computing the Variance Inflation Factor (VIF) for each predictor:

$$VIF_j = \frac{1}{1 - R_j^2}$$

where R_j^2 is the coefficient of determination when regressing $\mathbf{x}_j$ on the other predictors. A high VIF (e.g., greater than 5 or 10) indicates strong multicollinearity.

The condition number of $\mathbf{X}^T\mathbf{X}$ is another indicator of multicollinearity. It is defined as:

$$\kappa = \frac{\lambda_{\max}}{\lambda_{\min}}$$

where $\lambda_{\max}$ and $\lambda_{\min}$ are the largest and smallest singular values (or eigenvalues) of $\mathbf{X}^T\mathbf{X}$. A condition number $\kappa > 30$ suggests severe multicollinearity.

0.2 Dealing with outliers/influential points

We are going to use the full set of data and compute the results of the multi-linear regression between the body density and the other parameters. Then, we will use the information we dispose on the hat matrix to detect the possible presence of leverage points or outliers. The presence of these points might influence the model and the associated statistics. That's why, to see if these points are influent in the model, we will compute the Cook's Distance on them. This distance highlights the points that significantly impact the model coefficients when removed (which are the influential points).
Cook's distance for an observation i is defined as:

$$D_i = \frac{\sum_{j=1}^{n}(\hat{Y}_j - \hat{Y}_{j(i)})^2}{p \cdot MSE}, \tag{1}$$

where:

- $\hat{Y}_j$ is the predicted value for observation j using all data points,

- $\hat{Y}_{j(i)}$ is the predicted value for observation j after removing point i,
- MSE is the mean squared error of the model.

We recall here that removing any data point must be explained explicitly.

In practice, an observation is considered influential if its Cook's distance exceeds a certain threshold, typically:

$$D_i > \frac{4}{n}, \tag{2}$$

where n is the total number of observations. We can then try to remove these influential points to enhance the accuracy of the model.

0.3 The model

We assume that our model follow first this equation:

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \cdots + \beta_p X_p + \varepsilon \tag{3}$$

which can be written is a more compact form:

$$Y = \mathbf{X}\beta + \varepsilon \tag{4}$$

where $\mathbf{X}_i$ is a column matrix in which all the data of one parameter are included. Here, we assume that $\boldsymbol{\varepsilon} \sim \mathcal{N}(\mathbf{0}, \sigma^2\mathbf{I})$. It represents the error terms of the multicolinearity model. We are assuming that the errors are of mean 0, and more importantly, uncorrelated and with constant variance. This assumption has to be strongly verified to justify the validity of our model. The purpose of this regression is to estimate as accurately as possible the values of the estimator of β knowing the observation matrix X.

We first estimate the value of β by the Ordinary Least Squares (OLS) method. β is supposed to minimize the ordinary least-squares function:

$$S(\beta) = \sum_{i=1}^{n} (y_i - \mathbf{x}_i^T \beta)^2$$

which can also be written in matrix form as:

$$S(\beta) = (\mathbf{y} - \mathbf{X}\beta)^T (\mathbf{y} - \mathbf{X}\beta).$$

The OLS estimator of β is then given by:

$$\hat{\beta} = (\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T\mathbf{y}.$$

We notice that the expression of β requires the invertibility of the matrix $\mathbf{X}^T\mathbf{X}$, which then requires that the rank of $\mathbf{X}$ be full. If a clear relationship occurs between the input parameters, then the estimator of β would not be unique.

It is important to analyze the expectation and the variance of our estimator. Indeed, an accurate estimator should be able to predict well the final output when a new set of data is given.

By a simple calculous, we compute these values.

Expectation:

$$\mathbb{E}[\hat{\boldsymbol{\beta}}] = \boldsymbol{\beta}.$$

Variance:

$$\mathrm{Var}(\hat{\boldsymbol{\beta}}) = \sigma^2(\mathbf{X}^T\mathbf{X})^{-1}.$$

We can add the fact that the variance strongly depends of the inverse of $\mathbf{X}^T\mathbf{X}$. Consequently, if multicolinearity occurs, the variance of our estimator skyrockets. As a consequence, our observator will be sensitive to very small changes in future observations, which affects the accuracy of it.

Distribution: Since β is linear in y and y follows a normal distribution, then:

$$\hat{\boldsymbol{\beta}} \sim \mathcal{N}\left(\boldsymbol{\beta}, \sigma^2(\mathbf{X}^T\mathbf{X})^{-1}\right).$$

We have now to analyze if this model is relevant of not. We then use the following parameters to evaluate the model.

We have the coefficient R^2 defined by:

$$R^2 = 1 - \frac{SS_{res}}{SS_{tot}} \tag{5}$$

with:

- $SS_{res} = \sum(y_i - \hat{y}_i)^2$ is the sum of squared residuals,
- $SS_{tot} = \sum(y_i - \bar{y})^2$ is the total sum of squares.

This coefficient measures how well the predictors explain the variability of the density. It ranges from 0 to 1, where 1 indicates that our model explains all the variability, and 0 means that it explains none. The more predictors we have, the bigger this coefficient will be. The problem of overfitting might then occur: we could think that the model is perfect since it has a large R^2 coefficient, but it struggles to predict new observations.

That's why the define the adjusted R^2:

$$R^2_{adj} = 1 - \left(\frac{(1 - R^2)(n-1)}{n-p-1}\right) \tag{6}$$

This parameter includes a penalty while increasing the number of predictors.

We will also use the F-statistic. This statistic tests whether at least one of the independent variables significantly explains the variation in the dependent variable. A higher F-statistic indicates that the model is significantly better than a model without predictors.

We will check the statistical significance of the individual coefficients in the model with the p-value. A small p-value (typically less than 0.05) suggests strong evidence against the null hypothesis, meaning the predictor variable significantly impacts the dependent variable, while a big p-value challenges the impact of the predictor.

Finally, all these coefficient might tell us the accuracy of this regression to predict density.

0.4 How can we improve the model ?

We can first try to remove some predictors, considering that they might be highly correlated and lead to multicollinearity. The decision of removing parameters can be done with the VIF values and the p-values.
One way to deal with multicolinearity is by implementing variable selection, such as backward selection.

Backward selection

Given our model:

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \cdots + \beta_p X_p + \epsilon$$

Backward selection seeks to find the subset of predictors $\{X_{j_1}, \ldots, X_{j_k}\}$ that minimizes the Akaike Information Criterion (AIC):

$$\text{AIC} = 2k - 2\ln(\hat{L})$$

where k is the number of predictors and $\hat{L}$ is the maximized likelihood of the model.

The AIC provides a principled approach to model selection by balancing goodness-of-fit with model simplicity: it penalizes complexity to avoid overfitting

The backward algorithm can be summarized here:

1. **Start with the full model** .
2. **For each remaining predictor not in the model:**
   - Fit a model by removing the predictor.
   - Evaluate the model's performance (using AIC here).
3. **Remove the predictor** that provides the least significant improvement.
4. **Repeat steps 2-3** until no further improvement is found or a stopping criterion is met.

We can note that backward selection is a greedy algorithm and is not returning the optimal result, since it might remove relevant variables later.

In this report, we will use the Principal Component Regression (PCR) too to deal with multicolinearity.

Principal Component Regression

PCR is a regression technique that combines Principal Component Analysis (PCA) with linear regression. It reduces the dimensionality of the predictor space using PCA before performing regression. PCA transforms the original predictors into a new set of uncorrelated variables called principal components. The principal components are linear combinations of the original predictors and are ordered by the amount of variance that they explain.

The principal components Z are obtained by projecting X onto the space spanned by the matrix of eigenvectors T:

$$Z = XT$$

Then, we regress y on the first k principal components Z_k, where $k \leq p$. The value of k can be determined by looking at the sum of explained variance of the principal components. We have then the following model :

$$y = Z_k \beta_k + \epsilon$$

where:

β_k is the estimator of the k first principal components,

The use of PCR deals with multicolinearity and might then enhance the variance of our estimator and the accuracy of our model. However, it will be harder to understand the new PCR model, as the predictors will be linear combinations of the initial ones.

Leverage point

Another important phenomenon to study is the presence of outliers or leverage points in our data set. We can detect those points with the hat matrix $\mathbf{H}$:

$$\mathbf{H} = \mathbf{X}(\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T.$$

The diagonal elements h_{ii} measure the influence of each observation, and the distance of a point to the ERVH:

$$h_{ii} = \mathbf{x}_i^T(\mathbf{X}^T\mathbf{X})^{-1}\mathbf{x}_i.$$

A point with a large h_{ii} has consequently high leverage. This point might then significantly influence the parameters of the estimators and the accuracy of the model.

0.5 Pre-analysis of data

Before analyzing the results and making a multiple linear regression, we will see if the model explained by equation (2) is accurate. For this, we are going to plot the residuals vs the fitted values and establish the QQ-plot. This plot might inform us on the potential constant variance or not in the model. In order to explain our values and interpret them, we need the assumption that the errors are clearly following a normal distribution.

If the QQ-plot or the plot of the residuals shows us that the variance is not constant, we will have to transform our data before starting the model.

Variable transformation

Variable transformation consists of transforming the data by using a mathematical function, in order to meet the assumptions of normality and constant variance. While some easy transformations exists, the most popular one is the Box-Cox transformation. It is defined as:

$$y^{(\lambda)} = \begin{cases} \frac{y^{\lambda}-1}{\lambda} & \text{if } \lambda \neq 0 \\ \ln(y) & \text{if } \lambda = 0 \end{cases}$$

where λ is the hyper-parameter to be estimated.

The optimal λ is found by maximizing the log-likelihood function:

$$L(\lambda) = -\frac{n}{2} \ln \left[\frac{1}{n} \sum_{i=1}^{n} \left(y_i^{(\lambda)} - \bar{y}^{(\lambda)} \right)^2 \right] + (\lambda - 1) \sum_{i=1}^{n} \ln y_i$$

where $\bar{y}^{(\lambda)}$ is the mean of transformed values.

For this, we compute the likelihood function for a range of λ values (typically -2 to 2).

Variable transformation will be used if necessary. We can add that using variable transformation gives us poor interpretation of the results, since the predictors will change of nature.

0.6 Bootstrap method on the final model

Finally, we will use the Bootstrap method in our last model to improve the accuracy of the coefficients of our estimators.

The main idea of bootstrapping is to generate multiple samples of the same size as the original sample by drawing with replacement. For each generated sample, the estimator of interest is recalculated, allowing us to obtain an empirical distribution of this estimator.

The bootstrap algorithm can be summarized as follows:

1. We have our sample of data of size n.
2. We generate B samples of size n by drawing with replacement from the original sample.
3. For each bootstrap sample, we compute the OLS estimator
4. We have then an empirical distribution of the estimator, which can be used to derive confidence intervals for example

From the bootstrap distribution $\{\hat{\beta}^1, ..., \hat{\beta}^B\}$, we obtain the point estimates as $\hat{\beta}_{boot} = \frac{1}{B}\sum_{b=1}^{B}\hat{\beta}^b$; the standard errors $\text{SE}_{boot}(\hat{\beta}_j) = \sqrt{\frac{1}{B-1}\sum_{b=1}^{B}(\hat{\beta}_j^b - \hat{\beta}_{boot,j})^2}$.
We obtain even the confidence intervals of the predictors :

$$CI_{1-\alpha} = [q_{\alpha/2}, q_{1-\alpha/2}]$$

where q are quantiles of bootstrap distribution

1 Results

The command "any(is.na(data))" on R returns False, which means that we don't any NaaN values on the dataset. Furthermore, all the parameters are in decent units. Our data is ready to be used.

1.1 Analysis of Multicolinearity

First, we try to detect if there is any relationship between two variables.

Through the heatmap (figure 1), we see that there is a huge positive correlation between hip and weight, thigh and hip, weight and chest, and abdomen and chest (coefficients $>$ 0.90). We might then be aware of a possible relationship between abdomen, chest and hip in the future.

We plotted these parameters to detect potential linearity in Figure 2 and Figure 3 below.

We notice that the points seem to follow a linear pattern, which confirms the fact that there might be a relationship between some predictors. We also notice that there is a point in the dataset that is 'far' from the rest of the data: this motivates us to detect the presence of outliers in the future.

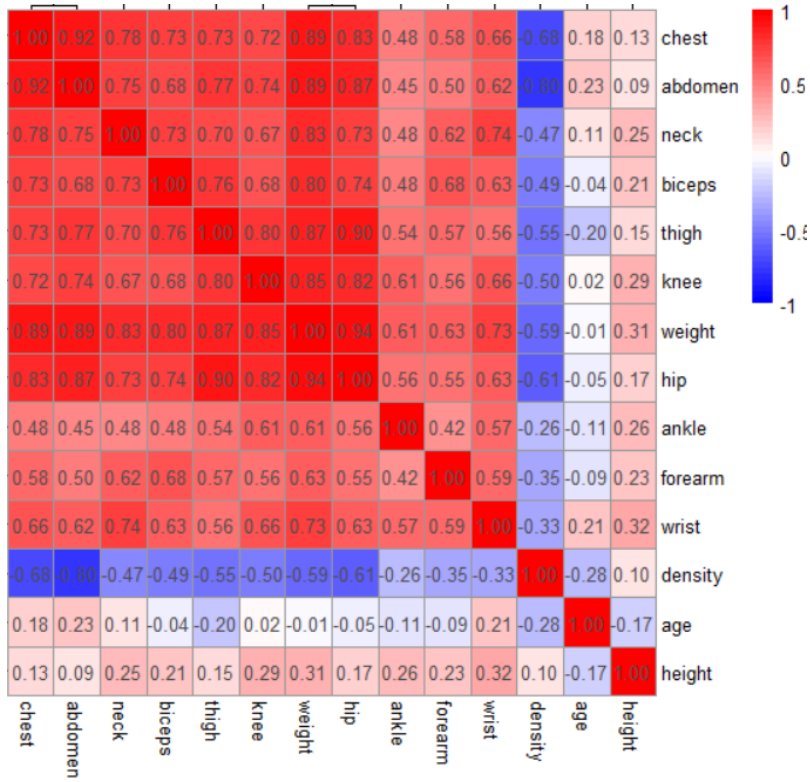

Figure 1: Heatmap of the parameters

However, as explained in the analysis section, the heatmap is clearly not sufficient to detect multicolinearity. In Figure 4, we displayed the values of the VIF Factors.

We have huge values for the predictors weight, hip, abdomen and chest (they are superior to 10). Furthermore, we have compared the eigenvalues of the model in Figure 5.

We see that the last eigenvalues are near zero. We have here:

$$\kappa = \frac{\lambda_{\max}}{\lambda_{\min}} \approx 300000000$$

which is huge.

As a consequence, we can strongly affirm that multicolinearity is likely present in the predictors. We will try to deal with this phenomenon later.

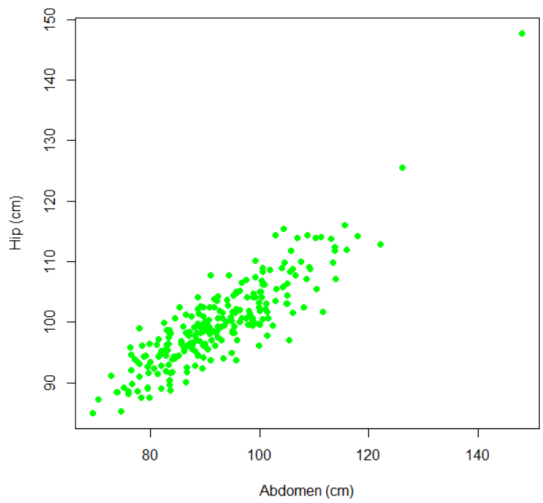

Figure 2: Hip vs Abdomen

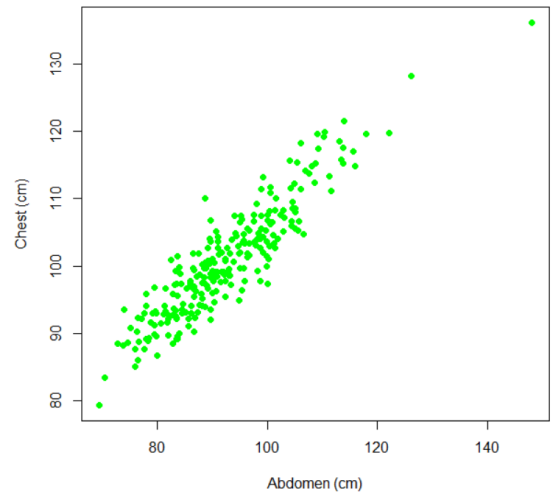

Figure 3: Chest vs Abdomen

```
   chest       age    weight    height      neck   abdomen       hip     thigh
9.460877  2.250450 33.509320  1.674591  4.324463 11.767073 14.796520  7.777865
    knee     ankle    biceps   forearm     wrist
4.612147  1.907961  3.619744  2.192492  3.377515
```

Figure 4: VIF Factors for each predictor

1.2 Validation of the model

We tried to see if the assumption of the errors following a normal law holds.

For this, we computed the QQ-plot. This graphic compares the quantiles of the observed data with the quantiles of the theoretical distribution.

We notice that the points approximately lie along the straight line composed of the identity function (which is what we want). We can also see a small deviation in the extremity of the graphic, which gives us the heavy tail shape here. This might be explained by the presence of some leverage points in the dataset, that we might try to remove next to enhance the model.

Then we tried to see if the residuals have constant variance. We plotted residuals vs fitted value in Figure 7 below.

Here, we don't notice any pattern which might think that variance increases or decreases. The QQ-plot and this graphic lead us to assume that the errors follow a normal law with a constant variance and zero mean: our assumptions made in the analysis are validated. We can then ensure that the F-statistic , the t-statistic and the confidence intervals are well defined.

```
 [1] 1.959280e+07 6.418672e+04 3.059850e+04 5.704342e+03 2.803950e+03
 [6] 1.934718e+03 1.030342e+03 6.376720e+02 5.280991e+02 4.318193e+02
[11] 3.763777e+02 2.723672e+02 6.345453e+01 6.157824e-02
```

Figure 5: Eigenvalues

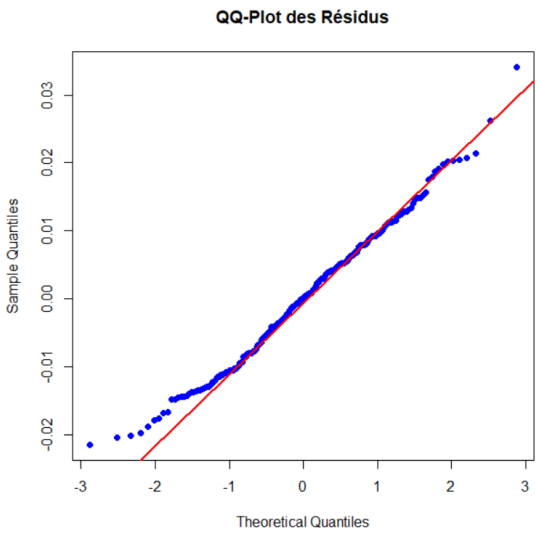

Figure 6: QQ-plot

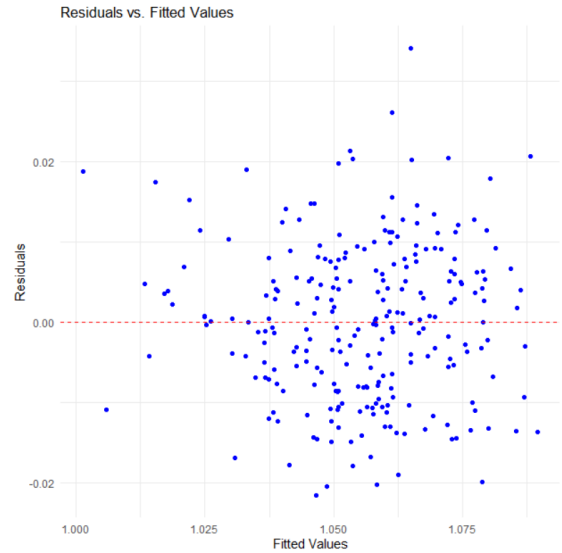

Figure 7: Residuals vs Fitted values

As a consequence, we are not going to do any variable transformation on our data. Indeed, since the assumptions are verified, we judge the transformation not useful here: we recall that by transforming some predictors, we lose interpretability.

2 A first computation of the linear model

We made a first model with all the possible dataset, without checking for outliers and multicolinearity. The results are presented in Figure 8:

We obtain a big F-statistic (51.6) and a small p-value (2.2 e-16), which highly confirms the fact that the model is relevant and might predict density. We reject therefore the hypothesis that all the coefficients of the estimators are null. Furthermore, we get a coefficient $R^2 = 0.74$, which is very satisfying here. The adjusted R^2 is quite the same since we don't have a lot of predictors here.

The different p-values for each coefficients show that there might be multicolinearity in the model. Indeed, a lot of p-values are mainly superior to 0.5, which causes to reevaluate their presence in the actual model. The main predictors that have huge p-values are: chest, knee, height, ankle, biceps, hip and age.

```
Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)  1.139e+00  4.030e-02  28.248  < 2e-16 ***
chest        1.232e-04  2.303e-04   0.535  0.59339
age         -1.203e-04  7.515e-05  -1.601  0.11062
weight       2.395e-04  1.243e-04   1.926  0.05528 .
height       1.498e-04  2.230e-04   0.672  0.50243
neck         1.075e-03  5.401e-04   1.991  0.04765 *
abdomen     -2.277e-03  2.008e-04 -11.335  < 2e-16 ***
hip          5.513e-04  3.390e-04   1.626  0.10521
thigh       -6.149e-04  3.354e-04  -1.833  0.06799 .
knee        -4.844e-05  5.622e-04  -0.086  0.93141
ankle       -6.314e-04  5.145e-04  -1.227  0.22094
biceps      -5.755e-04  3.976e-04  -1.448  0.14907
forearm     -1.017e-03  4.626e-04  -2.198  0.02891 *
wrist        3.959e-03  1.243e-03   3.185  0.00164 **
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.01 on 238 degrees of freedom
Multiple R-squared:  0.7381,    Adjusted R-squared:  0.7238
F-statistic:  51.6 on 13 and 238 DF,  p-value: < 2.2e-16
```

Figure 8: Result of the basic model

Finally, we notice that the coefficients of the estimator have quite the same magnitude. This discourage us to use the LASSO estimate to deal with multicolinearity, since this estimator selections the parameters with values above a definite threshold.

3 Backward elimination

We have already seen that some variables are highly correlated, which causes multicolinearity in the model. We will then try to remove some predictors to avoid overfitting and to narrow the relationships between some predictors.

```
Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)  1.149e+00  2.730e-02  42.082  < 2e-16 ***
age         -1.206e-04  7.181e-05  -1.680  0.09429 .
weight       2.587e-04  9.477e-05   2.730  0.00679 **
neck         1.173e-03  5.261e-04   2.229  0.02670 *
abdomen     -2.219e-03  1.677e-04 -13.227  < 2e-16 ***
hip          4.887e-04  3.235e-04   1.511  0.13219
thigh       -7.050e-04  3.111e-04  -2.266  0.02435 *
biceps      -5.451e-04  3.918e-04  -1.391  0.16540
forearm     -9.972e-04  4.559e-04  -2.187  0.02969 *
wrist        3.603e-03  1.186e-03   3.038  0.00264 **
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.009967 on 242 degrees of freedom
Multiple R-squared:  0.7355,    Adjusted R-squared:  0.7257
F-statistic: 74.79 on 9 and 242 DF,  p-value: < 2.2e-16
```

Figure 9: New model using backward elimination

Backward elimination has removed the predictors "chest", "height", "knee" and "ankle" from the model. By removing them, we highly improve the AIC coefficient: from 2307 to 1595. We notice that the predictors removed were the ones with the highest p-values,

which is very satisfying. By using this backward elimination, we are then able to reduce multicolinearity and to improve the F-statistic (from 51.6 to 74.79).

4 Model without influential points

Now, we are going to analyze the presence of influential points that might affect our model significantly. We use for this the Cook's distance. We decide here to remove all the points that are above the threshold of

$$D_i > \frac{4}{n}, \tag{7}$$

We plotted the influential points in Figure 9.

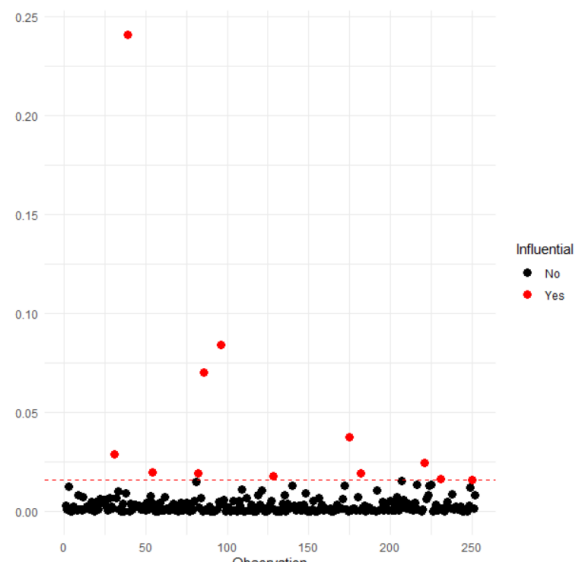

Figure 10: Cook's Distance and detection of influential points

We then made the new model without these points (Figure 10):

```
Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)  1.113e+00  4.039e-02  27.559  < 2e-16 ***
chest        2.351e-04  2.248e-04   1.046  0.29683
age         -1.350e-04  7.265e-05  -1.858  0.06450 .
weight       1.112e-04  1.279e-04   0.870  0.38536
height       1.662e-04  2.118e-04   0.785  0.43348
neck         1.223e-03  5.273e-04   2.319  0.02129 *
abdomen     -2.263e-03  1.932e-04 -11.711  < 2e-16 ***
hip          6.217e-04  3.251e-04   1.912  0.05710 .
thigh       -5.604e-04  3.213e-04  -1.744  0.08245 .
knee         1.899e-04  5.426e-04   0.350  0.72672
ankle       -1.689e-04  8.108e-04  -0.208  0.83511
biceps      -4.601e-04  4.123e-04  -1.116  0.26569
forearm     -9.348e-04  5.339e-04  -1.751  0.08133 .
wrist        3.586e-03  1.263e-03   2.839  0.00494 **
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.009237 on 226 degrees of freedom
Multiple R-squared:  0.7645,    Adjusted R-squared:  0.7509
F-statistic: 56.43 on 13 and 226 DF,  p-value: < 2.2e-16
```

Figure 11: New model without influential points

We notice that all the statistics have been enhanced due to the suppression of the influential points. Our model better expresses the variability of the data(increased coefficient R^2 from 0.73 to 0.76). The global F-statistic is higher as well. Furthermore, the p-values of each predictor have decreased. However, some p-values still remain too large, which can be solved by PCR.

5 Principal Component Regression

We will do a PCR analysis to improve the model and try to reduce multicolinearity as much as possible First, we compute the variance explained by each principal component (Figure 11 and Figure 12).

We see that the first principle component already takes 60 percent of the variance. The first six principal component of the analysis are explaining more than 90 percent of the variance. Thus, we choose these six components as principal components to make the regression. We used cross validation on 10 sets to enhance the results.

We see that (Figure 14), as mentioned above, the six first principal components explain more than 90 percent of the explained variance. Moreover, they explain 65 percent of the variability in density, which is a reasonable result. Finally, the coefficients have a much larger weight than with the ordinary least-square estimation: this is due to the linear combination, and the fact that we have less coefficient now. The coefficient R^2 is smaller than the previous one (0.65 compared to 0.76): we expected this to happen since R^2 obtained by PCR should be less than equal to the one given by the full model. We see that we could just take the five first principal components, as the coefficient from the PCR model with 5 components and the one with 6 are quite the same.

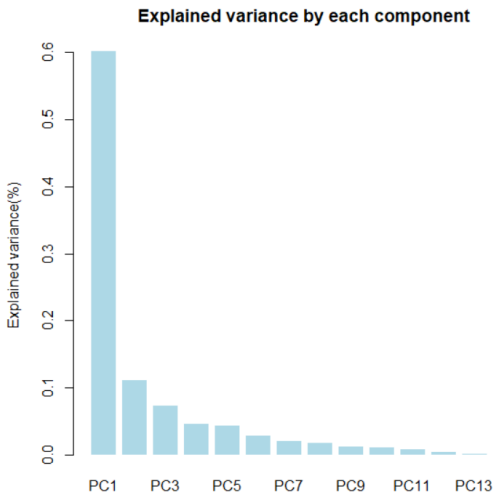

Figure 12: Explained Variance

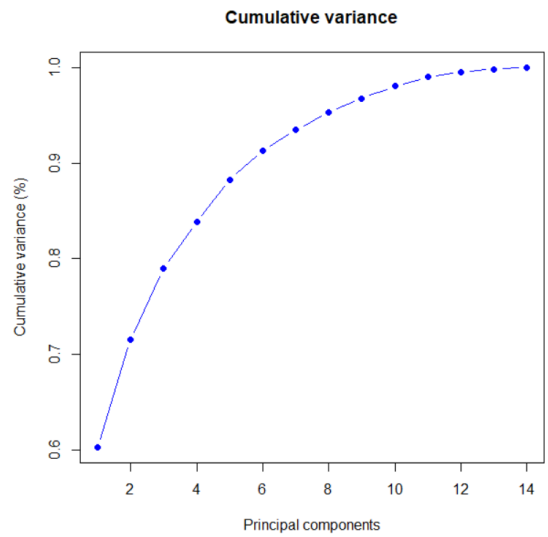

Figure 13: Cumulative Variance

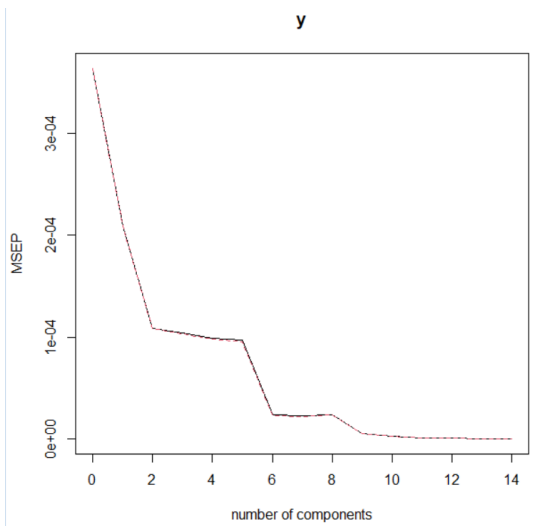

Figure 14: MSEP per principal component

```
Number of components considered: 6

VALIDATION: RMSEP
Cross-validated using 10 random segments.
       (Intercept)  1 comps  2 comps  3 comps  4 comps  5 comps  6 comps
CV         0.01855  0.01466  0.01285  0.01262  0.01197  0.01146  0.01148
adjCV      0.01855  0.01464  0.01284  0.01257  0.01188  0.01142  0.01143

TRAINING: % variance explained
          1 comps  2 comps  3 comps  4 comps  5 comps  6 comps
X           62.76    73.43    81.39    86.12    90.37    92.70
density     38.40    52.90    58.87    61.77    65.95    65.95
```

Figure 15: Result of the PCR

Final model

We will now choose the final model. We decided to take the model made with the bootstrap method with the influential points removed. The Principal Component Regression might then be more accurate to predict density, but we loose the interpretation of the predictors

```
       (Intercept)  1 comps  2 comps  3 comps  4 comps  5 comps  6 comps
train     0.000000   0.3840   0.5290   0.5887   0.6177   0.6595   0.6595
CV       -0.008386   0.3702   0.5158   0.5331   0.5803   0.6151   0.6138
```

Figure 16: R^2 coefficient with CV for the PCR

for future observations. The backward elimination was able to remove some predictors that caused multicolinearity, and to simplify the model without losing interpretation. Here is a summary of the backward elimination without the influential points:

```
Residuals:
       Min         1Q     Median         3Q        Max
-0.0223061 -0.0064269  0.0002661  0.0061125  0.0251260

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)  1.1432137  0.0260363  43.908  < 2e-16 ***
age         -0.0001279  0.0000681  -1.878  0.06163 .
weight       0.0001973  0.0000915   2.157  0.03206 *
neck         0.0012480  0.0005096   2.449  0.01507 *
abdomen     -0.0022041  0.0001629 -13.528  < 2e-16 ***
hip          0.0005432  0.0003105   1.749  0.08156 .
thigh       -0.0006378  0.0002951  -2.161  0.03173 *
biceps      -0.0004418  0.0004010  -1.102  0.27174
forearm     -0.0009069  0.0005268  -1.722  0.08649 .
wrist        0.0034506  0.0011471   3.008  0.00292 **
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.009187 on 230 degrees of freedom
Multiple R-squared:  0.7629,    Adjusted R-squared:  0.7537
F-statistic: 82.25 on 9 and 230 DF,  p-value: < 2.2e-16
```

Figure 17: Final model

Thanks to our computations, we have improved the F-statistic, the R^2 coefficient, and the residual standard error. Furthermore, the p-values per coefficient seem to be more reasonable (except for the predictor biceps, which is still too large).

We now did the bootstrap method on 1000 iterations to compute the confidence intervals of the coefficient of the predictors, and to ensure more accuracy. The results are presented in Figure 18.

Finally, we used 10 folds cross validation to have a more accurate R^2 coefficient. This method gave us the value $R^2 = 0.76$ with standard deviation of 0.06. The combination of the statistic coefficient gives us, at the end, a very satisfying model.

	Coefficient (β)	IC bootstrap 2.5%	IC bootstrap 97.5%
age	-0.0001279087	-2.414584e-04	-1.449155e-05
weight	0.0001973550	2.875013e-05	3.690595e-04
neck	0.0012480449	3.092408e-04	2.201125e-03
abdomen	-0.0022040862	-2.475653e-03	-1.896276e-03
hip	0.0005431680	-1.128146e-05	1.067636e-03
thigh	-0.0006378088	-1.172959e-03	-1.277172e-04
biceps	-0.0004417655	-1.178741e-03	2.732333e-04
forearm	-0.0009068585	-1.780322e-03	9.342947e-05
wrist	0.0034505902	1.244973e-03	5.562670e-03

Figure 18: Coefficients and confidence intervals for the predictors

Conclusion

The regression between density and the other parameters is accurate and might lead to interesting results: the statistics and coefficient show that some predictors are well-suited to predict density, and the data exhibit constant variance for the residuals. Some points in the data set are considered as influent, and removing these points help to have a more accurate model. A more detailed discussion on whether to exclude these points from the dataset may be necessary. However, if we want to predict new observations, it might be interesting to do so. Finally, the variance of the estimator was too high, due to multicolinearity, which could lead to poor prediction for future data points. We would like to avoid this problem. To address this issue, we did the PCR analysis and some variable selection with the backward elimination. The final model chosen has a lower variance, and consequently deals with multicolinearity without losing interpretation. Considering the values of some coefficient, we can affirm that this model may provide reliable predictions for future new observations.